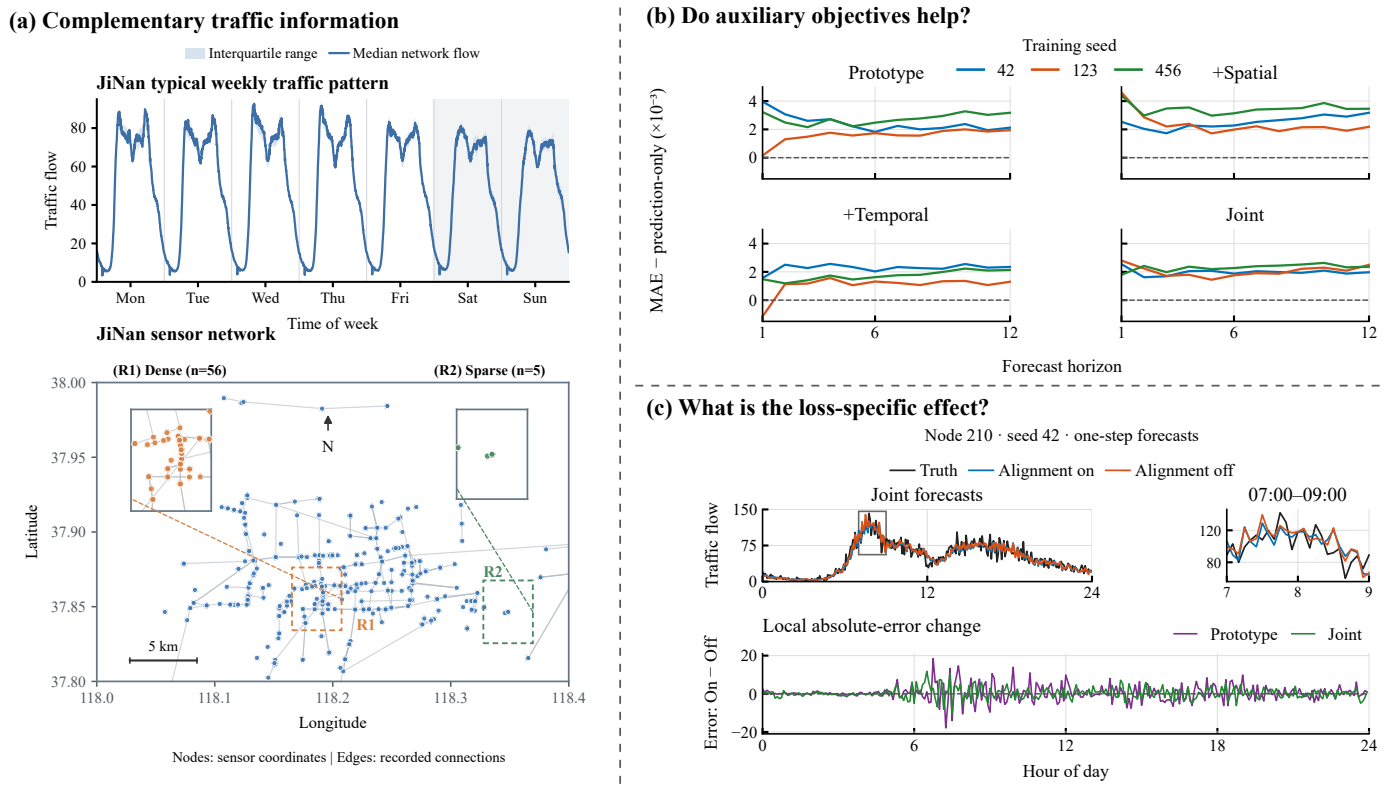

(a) Complementary traffic information
Interquartile range
Median network flow
JiNan typical weekly traffic pattern
Traffic flow
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Time of week
JiNan sensor network
(R1) Dense (n=56)
(R2) Sparse (n=5)
N
R1
R2
5 km
Latitude
Longitude
Nodes: sensor coordinates | Edges: recorded connections
(b) Do auxiliary objectives help?
Training seed
Prototype
42
123
456
+Spatial
+Temporal
Joint
MAE − prediction-only (×10⁻³)
Forecast horizon
(c) What is the loss-specific effect?
Node 210 · seed 42 · one-step forecasts
Truth
Alignment on
Alignment off
Joint forecasts
07:00–09:00
Traffic flow
Local absolute-error change
Prototype
Joint
Error: On − Off
Hour of day